

ROLLING HILLS AGCM, AB, Canada

WMO: 711050

Lat: 50.2650N

Lon: 111.7011W

Elev: 2116

StdP: 13.61

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.4	-17.0	-29.4	1.1	-23.2	-22.7	1.7	-16.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.2	89.6	62.5	86.0	61.8	82.7	61.1	65.9	82.3	64.1	80.7	62.6	79.1	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	84.3	72.2	58.1	78.0	69.6	56.3	73.1	67.5	31.8	82.2	30.4	80.8	29.2	79.3	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-33.7	96.1	5.4	3.0	-37.6	98.3	-40.8	100.1	-43.8	101.8	-47.8	104.0		
(5)			-33.9	70.0	5.4	1.9	-37.8	71.4	-40.9	72.4	-44.0	73.5	-47.9	74.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	40.6	17.4	15.8	29.0	41.1	52.0	59.7	65.4	64.1	55.3	42.0	28.1	15.3	(6)
(7)		DBStd	21.41	15.96	15.79	14.67	8.87	7.76	5.80	4.95	5.74	8.35	9.36	13.81	15.87	(7)
(8)		HDD50	5031	1010	957	651	285	69	4	0	1	41	276	661	1076	(8)
(9)		HDD65	9073	1474	1376	1115	717	406	177	56	86	305	712	1108	1541	(9)
(10)		CDD50	1596	0	0	1	18	132	295	478	439	200	30	3	0	(10)
(11)		CDD65	164	0	0	0	0	3	18	69	60	14	0	0	0	(11)
(12)		CDH74	3873	0	0	0	22	227	470	1387	1285	464	18	0	0	(12)
(13)		CDH80	1319	0	0	0	4	49	127	488	487	163	1	0	0	(13)
(14)	Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15)	Precipitation	PrecAvg	12.9	0.6	0.4	0.6	1.0	1.7	2.8	1.6	1.2	0.6	0.6	0.5	(15)	
(16)		PrecMax	19.1	1.2	0.8	1.5	2.8	3.8	7.3	4.4	3.0	3.3	1.7	1.6	1.1	(16)
(17)		PrecMin	6.1	0.2	0.1	0.0	0.1	0.2	0.6	0.2	0.1	0.2	0.1	0.1	0.1	(17)
(18)		PrecStd	2.7	0.3	0.2	0.4	0.6	0.9	1.5	1.1	0.7	0.8	0.4	0.4	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.3	52.2	64.6	77.1	85.2	89.8	93.2	93.8	90.6	76.8	65.4	49.9	(19)
(20)		MCWB	41.5	41.3	47.4	52.9	57.8	63.3	64.8	63.9	60.5	56.0	49.5	40.5	(20)	
(21)		2%	DB	46.5	46.1	59.8	69.9	80.3	83.5	89.1	89.5	85.2	69.3	56.9	44.5	(21)
(22)		MCWB	38.7	37.8	45.2	50.0	56.5	60.8	64.1	62.1	59.0	51.9	45.1	37.7	(22)	
(23)		5%	DB	42.7	41.3	54.7	64.3	75.6	79.2	85.3	85.6	79.4	64.1	51.9	40.8	(23)
(24)		MCWB	36.8	35.0	42.3	46.5	54.3	59.4	63.1	61.8	57.1	49.4	42.2	35.4	(24)	
(25)		10%	DB	39.3	37.3	48.8	59.3	70.6	75.5	82.0	81.6	73.9	59.0	46.9	37.6	(25)
(26)		MCWB	34.6	33.0	39.3	43.9	52.2	57.7	62.3	60.8	55.1	46.6	38.9	32.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.1	42.7	48.5	53.1	59.6	66.0	69.2	67.0	62.7	56.7	49.7	41.8	(27)
(28)		MCDWB	49.9	52.0	62.1	72.6	80.9	82.2	84.7	83.6	83.5	72.4	63.7	48.8	(28)	
(29)		2%	WB	39.2	38.1	46.0	50.4	57.1	63.4	66.7	64.8	60.1	53.1	45.9	38.1	(29)
(30)		MCDWB	45.8	45.2	58.1	68.3	77.1	79.6	81.8	82.9	80.2	66.7	55.8	43.9	(30)	
(31)		5%	WB	36.9	35.6	42.8	47.5	55.5	61.0	65.0	63.2	58.1	50.0	42.7	35.7	(31)
(32)		MCDWB	42.3	40.9	53.5	61.6	72.4	75.7	80.4	81.0	77.1	62.9	51.0	40.8	(32)	
(33)		10%	WB	34.8	32.8	39.6	44.9	53.4	58.9	63.4	61.6	55.9	47.5	39.4	33.1	(33)
(34)		MCDWB	39.4	37.0	48.4	57.9	67.9	72.6	78.9	78.5	72.4	58.0	46.0	37.3	(34)	
(35)	Mean Daily Temperature Range	MDBR	19.7	21.7	23.3	26.5	27.2	25.6	29.2	29.8	29.1	25.9	21.8	19.4	(35)	
(36)		5% DB	MCDBR	20.2	25.4	31.0	38.2	36.6	33.3	35.9	38.3	40.2	33.3	28.2	21.1	(36)
(37)		MCWBR	14.3	18.2	18.2	19.4	16.0	14.4	15.2	15.2	17.3	17.5	17.4	15.6	(37)	
(38)		5% WB	MCDBR	20.3	24.6	30.1	34.6	32.2	30.7	32.2	33.8	37.5	32.3	27.7	21.3	(38)
(39)		MCWBR	14.8	18.1	18.1	18.2	14.4	14.5	14.9	14.3	16.3	17.5	17.9	16.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.277	0.282	0.310	0.364	0.384	0.372	0.371	0.397	0.347	0.305	0.276	0.264	(40)	
(41)		taud	2.382	2.441	2.417	2.262	2.259	2.349	2.370	2.267	2.393	2.487	2.464	2.383	(41)	
(42)		Ebn at Noon	245	276	286	278	277	281	279	265	268	263	246	232	(42)	
(43)		Edh at Noon	20	24	30	39	41	38	36	38	30	23	18	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	384	697	1126	1560	1854	2006	2182	1758	1258	775	428	297	(44)	
(45)		RadStd	37	61	79	109	128	121	107	112	106	82	35	27	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page